

**PIMPRI CHINCHWAD EDUCATION TRUST'S
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
SECTOR NO.26, PRADHIKARAN, NIGDI, PUNE-
411044**



MDM CAPSTONE PROJECT SYNOPSIS

Project Group ID: Grp_Cap_63

***Title : Development of an Intelligent Retrieval-Augmented Generation (RAG) Chatbot
Using Open-Source Large Language Models***

Project Group Members:

123B1B072	Adhyayan Goyal
123B1B075	Aditya Chavan
123B1B082	Harsh Asalkar
123B1B100	Likhith Chaudhary

**Project Guide:
Dr. Jyoti Kulkarni**

**Head of the Department:
Prof. Dr. Anuradha Thakare**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI & ML)
ACADEMIC YEAR 2026 - 2027**

Table of Contents

Abstract	2
1. Introduction.....	3
2. Background.....	3
2.1 Problem Statement	3
2.2 Motivation / Need of the Project	4
2.3 Objectives	4
2.4 Scope of the Project.....	4
3. Literature survey	5
3.1 Summary of Literature Survey.....	6
4. Methodology	6
4.1 Block Diagram	6
4.2 Hardware Requirements	6
4.3 Software Requirements	7
4.4 Flowchart	7
4.5 Working Principle	8
5. Expected Conclusion	8
6. References.....	8

Abstract

Large Language Models (LLMs) have transformed natural language understanding and generation; however, they often suffer from limitations such as hallucination, outdated knowledge, and the inability to access domain-specific information. Retrieval-Augmented Generation (RAG) addresses these challenges by combining semantic retrieval with generative AI, enabling responses to be grounded in external knowledge sources rather than relying solely on the model's internal parameters. This project proposes the development of an intelligent document-based chatbot that integrates a RAG pipeline with an open-source Large Language Model such as Gemma or Llama. The system will process user-uploaded documents, generate vector embeddings, store them in a vector database, retrieve the most relevant information using semantic search, and utilize the retrieved context to generate accurate, context-aware, and explainable responses. The proposed solution aims to reduce hallucinations, improve factual accuracy, provide source-backed answers, and enable efficient interaction with large collections of documents. Owing to its modular and scalable architecture, the system can be applied in domains such as education, enterprise knowledge management, customer support, research, and healthcare while maintaining flexibility through the use of open-source technologies.

1. Introduction

Large Language Models (LLMs) have revolutionized Natural Language Processing by enabling machines to understand and generate human-like text. However, these models possess static knowledge limited to their training data and may generate inaccurate information when queried about recent or domain-specific topics.

Retrieval-Augmented Generation (RAG) enhances LLM performance by combining semantic search with language generation. Instead of relying entirely on the model's internal knowledge, relevant information is retrieved from a document repository and incorporated into the prompt before generating a response.

The proposed project aims to build a web-based chatbot capable of answering user questions based on uploaded documents. By integrating semantic search, vector databases, and open-source language models, the chatbot will provide reliable, context-aware, and explainable responses.

2. Background

2.1 Problem Statement

Existing AI chatbots frequently produce inaccurate or fabricated responses because they rely only on pre-trained knowledge. They cannot directly access organization-specific documents or newly added information without retraining. This limits their usefulness in domains where accurate, document-grounded information is essential.

The challenge is to design an intelligent chatbot capable of retrieving relevant information from custom knowledge bases and generating precise responses while minimizing hallucinations.

2.2 Motivation / Need of the Project

Organizations increasingly rely on digital documentation, including policies, reports, manuals, research papers, and technical documentation. Searching through these documents manually is time-consuming and inefficient.

A RAG-based chatbot enables users to interact with these documents conversationally, reducing search time and improving information accessibility. Using open-source LLMs also minimizes deployment costs while ensuring data privacy.

2.3 Objectives

- Develop an intelligent document-based chatbot.
- Integrate Retrieval-Augmented Generation (RAG) architecture.
- Use an open-source Large Language Model for response generation.
- Convert uploaded documents into vector embeddings.
- Retrieve context using semantic similarity search.
- Generate accurate, context-aware responses.
- Reduce hallucinations compared to traditional LLM chatbots.
- Provide references to retrieved documents.

2.4 Scope of the Project

The project focuses on building a web-based conversational AI system capable of answering questions from uploaded documents.

The proposed system supports:

- PDF document processing
- Semantic document retrieval
- Vector database storage
- Open-source LLM integration
- Multi-document querying
- Context-aware conversations
- Source citation

Future extensions may include OCR support, multilingual interaction, voice assistance, cloud deployment, and image understanding.

3. Literature survey

Sr. No	Title of the Paper	Year	Publisher	Methodology	Conclusion
1	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	2020	NeurIPS	Combined Dense Passage Retrieval (DPR) with a seq2seq generator to retrieve external knowledge before response generation.	Introduced the original RAG architecture, significantly improving factual accuracy and reducing hallucinations in knowledge-intensive tasks. (ML Anthology)
2	Dense Passage Retrieval for Open-Domain Question Answering	2020	EMNLP	Used dual-encoder dense embeddings instead of keyword-based retrieval for semantic document search.	Demonstrated that dense retrieval substantially outperforms traditional BM25 retrieval, making it ideal for RAG pipelines. (arXiv)
3	REALM: Retrieval-Augmented Language Model Pre-Training	2020	ICML	Integrated retrieval during language model pre-training using a differentiable retriever and external knowledge base.	Showed that retrieval-enhanced language models improve question answering performance while providing interpretable knowledge access. (Reddit)
4	FiD: Leveraging Passage Retrieval with Generative Models for Open-Domain Question Answering	2021	ACL	Encoded multiple retrieved passages independently before fusing them during generation.	Improved answer quality by effectively utilizing multiple retrieved documents, influencing modern multi-document RAG systems.
5	Atlas: Few-shot Learning with Retrieval-Augmented Language Models	2022	ICML	Combined retrieval with few-shot learning to improve performance without extensive task-specific fine-tuning.	Demonstrated that retrieval enables strong performance even with limited training examples, improving scalability of RAG-based systems.
6	Retrieval-Augmented Generation for Large Language Models: A Survey	2023	arXiv Survey	Reviewed retrieval, indexing, embedding, augmentation, evaluation metrics, and advanced RAG architectures.	Provides a comprehensive overview of modern RAG architectures, implementation strategies, challenges, and future research directions. (arXiv)

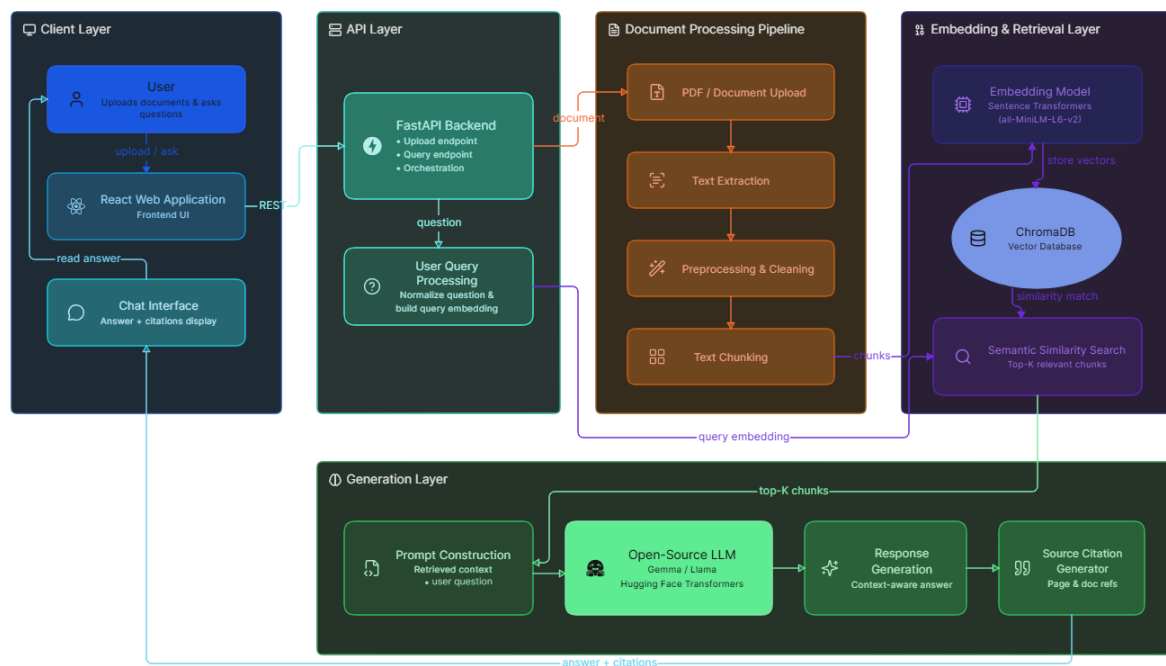
3.1 Summary of Literature Survey

The literature consistently demonstrates that Retrieval-Augmented Generation significantly enhances the factual reliability, transparency, and adaptability of Large Language Models by integrating external knowledge retrieval into the generation process. Foundational works such as RAG, DPR, and REALM establish the theoretical basis for combining semantic retrieval with language generation, while later architectures such as FiD and Atlas improve multi-document reasoning and retrieval efficiency. Recent survey papers further highlight the evolution of RAG techniques, including advanced retrieval strategies, prompt augmentation, evaluation metrics, and scalable deployment architectures. Collectively, these studies indicate that RAG-based systems provide a practical and efficient solution for building domain-specific conversational AI with reduced hallucinations and improved explainability, making them highly suitable for enterprise and educational applications.

4. Methodology

4.1 Block Diagram

Intelligent RAG Chatbot Using Open-Source LLMs — System Architecture



4.2 Hardware Requirements

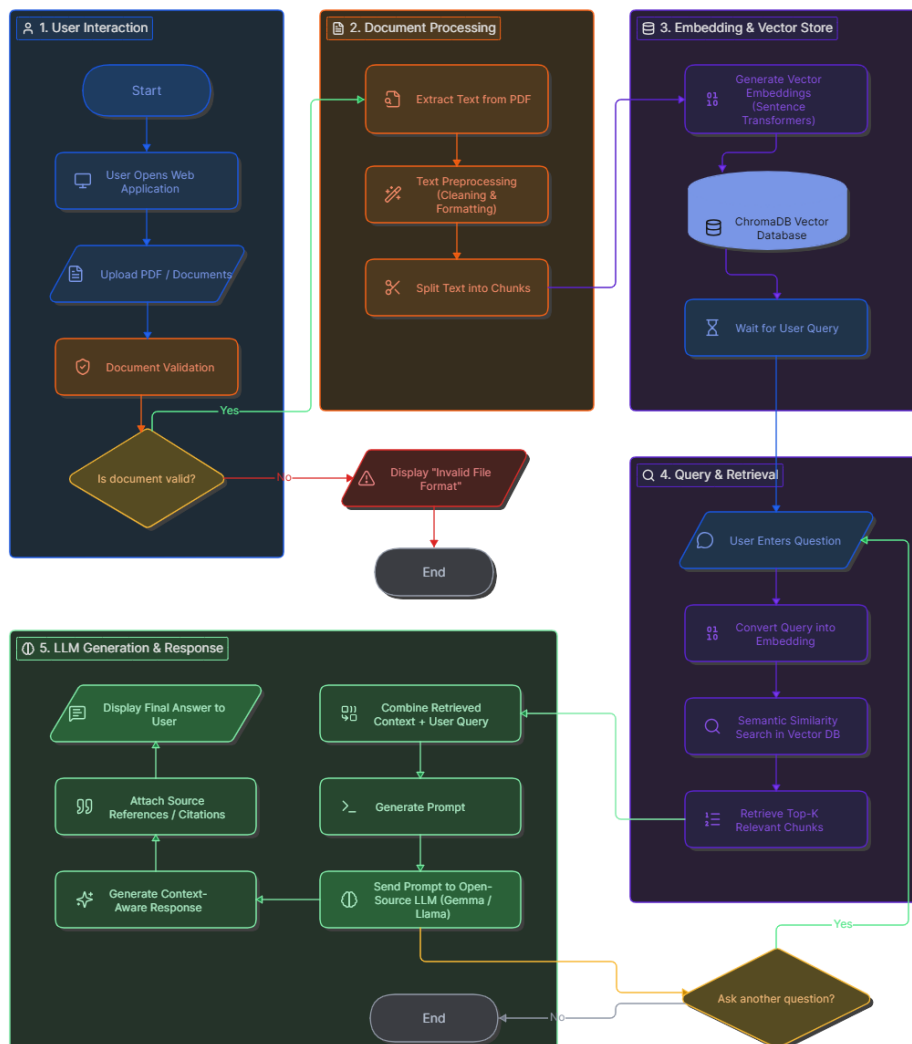
- Intel Core i5 Processor or higher
- 8 GB RAM (Minimum)
- 16 GB RAM (Recommended)
- 256 GB Storage
- Stable Internet Connection

4.3 Software Requirements

- Windows/Linux
- Python 3.11+
- FastAPI
- React.js
- LangChain
- ChromaDB
- HuggingFace Transformers
- Sentence Transformers
- Git
- VS Code

4.4 Flowchart

Development of an Intelligent Retrieval-Augmented Generation (RAG) Chatbot
Using Open-Source Large Language Models



4.5 Working Principle

The system begins by accepting one or more documents from the user. The uploaded documents are processed to extract text, which is then divided into smaller chunks. Each chunk is converted into a numerical vector representation using an embedding model and stored in a vector database.

When the user submits a query, the query is converted into an embedding and compared against the stored document vectors using semantic similarity search. The most relevant document chunks are retrieved and combined with the user's question to create an enhanced prompt. This prompt is provided to an open-source language model, which generates a response grounded in the retrieved context. The chatbot then presents the answer along with references to the source documents.

5. Expected conclusions

The proposed system is expected to demonstrate the effectiveness of Retrieval-Augmented Generation in improving chatbot performance. Compared to conventional chatbots, the developed system should provide more accurate, context-aware, and trustworthy responses while reducing hallucinations. The integration of semantic search and open-source language models is anticipated to enable efficient retrieval of relevant information from uploaded documents. The project will also showcase the practical application of modern Generative AI techniques in building scalable and cost-effective conversational systems suitable for educational, enterprise, and research environments.

6. References

Research Papers

1. Lewis, P., et al. (2021). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS. [<https://arxiv.org/pdf/2005.11401>]
2. Karpukhin, V., et al. (2020). *Dense Passage Retrieval for Open-Domain Question Answering*. EMNLP. [<https://arxiv.org/pdf/2004.04906>]
3. Guu, K., et al. (2020). *REALM: Retrieval-Augmented Language Model Pre-Training*. ICML. [<https://dl.acm.org/doi/abs/10.5555/3524938.3525306>]
4. Izacard, G., & Grave, E. (2021). *Leveraging Passage Retrieval with Generative Models for Open-Domain Question Answering (FiD)*. ACL. [<https://aclanthology.org/2021.eacl-main.74/>]
5. Izacard, G., et al. (2022). *Atlas: Few-shot Learning with Retrieval Augmented Language Models*. ICML. [<https://dl.acm.org/doi/epdf/10.5555/3648699.3648950>]
6. Gao, Y., et al. (2023). *Retrieval-Augmented Generation for Large Language Models: A Survey*. [<https://www.rivista.ai/wp-content/uploads/2025/09/2312.10997v5.pdf>]

Technical Documentation

- [LangChain Documentation](#)
- [ChromaDB Documentation](#)
- [Hugging Face Transformers Documentation](#)
- [Sentence Transformers Documentation](#)
- [OpenSearch RAG Chatbot Tutorial](#)